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SIMATS Online Programming Lab

C++ PROGRAMMING

CO2_AT2_Scenario Based Assessment

View Only

QUESTIONS

Q1

AI-Based Student Performance Analytics System

10 marks

✓

Q2

Digital Healthcare Patient Monitoring System

10 marks

✓

Q3

Digital Wallet and FinTech Transaction System

10 marks

✓

Q4

Smart Agriculture Crop Monitoring System

10 marks

✓

Q5

Smart City IoT Energy Management System

10 marks

✓

Q1 AI-Based Student Performance Analytics System

10 marks

AI-Based Student Performance Analytics System ♦ Scenario: Intelligent Education. A university wants to develop an AI-assisted Student Performance Analytics System to monitor student academic performance and identify students who may need additional academic support.

A. Create a class Student with attributes such as Student ID, Name, Marks in different subjects, Average, and Performance Level.

B. Implement member functions to input student details, calculate average marks, determine performance level, and display the student report.

C. Use an array of objects to manage records of multiple students.

D. Include a static member to maintain the total number of students analyzed.

E. Apply proper encapsulation using private data members and public member functions.

Your Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Student {
5 private:
6     int id, m1, m2, m3;
7     string name, level;
8     float avg;
9 public:
```

Input

```
3
101 Ravi 80 75 90
102 Kumar 60 55 70
103 Priya 40 45 35
```

Output

hp

Q2 Digital Healthcare Patient Monitoring System

Q1 
AI-Based Student Performance
Analytics System
10 marks

Q2 
Digital Healthcare Patient
Monitoring System
10 marks

Q3 
Digital Wallet and FinTech
Transaction System
10 marks

Q4 
Smart Agriculture Crop
Monitoring System
10 marks

Q5 
Smart City IoT Energy
Management System
10 marks

5/5 answered

Digital Healthcare Patient Monitoring System ♦ Scenario: Smart Healthcare. A modern hospital wants a Patient Monitoring System that maintains patient information and automatically calculates basic health-risk parameters.

- A. Design a class Patient with attributes such as Patient ID, Name, Age, Heart Rate, Blood Pressure, and Health Status.
- B. Implement member functions to input patient details, calculate/identify the health status based on patient report.
- C. Store multiple patient records using an array of objects.
- D. Use a static member to track the total number of patients monitored.
- E. Demonstrate data hiding, encapsulation, and structured object-oriented program design.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Patient {
4 private:
5     int id, age, heartRate, bp;
6     string name, healthStatus;
7 public:
8     static int count;
9     void input() {
10         cout << "Patient ID: ";
11         cin >> id;
12         cout << "Name: ";
13         cin >> name;
```

Input

2
101
John
25

Output

Q3 Digital Wallet and FinTech Transaction System

Q1
AI-Based Student Performance
Analytics System
10 marks

Q2
Digital Healthcare Patient
Monitoring System
10 marks

Q3
Digital Wallet and FinTech
Transaction System
10 marks

Q4
Smart Agriculture Crop
Monitoring System
10 marks

Q5
Smart City IoT Energy
Management System
10 marks

5/5 answered

Digital Wallet and FinTech Transaction System ♦ Scenario: Cashless Economy. A FinTech company wants to develop a Management System to maintain customer wallet information and perform secure financial transactions.

- A. Create a class DigitalWallet with attributes such as Wallet ID, Customer Name, Mobile Number, and Balance.
- B. Implement member functions for wallet creation, money deposit, money withdrawal, balance enquiry, and transaction.
- C. Use an array of objects to manage multiple customer wallets.
- D. Include a static member to maintain the total number of active wallets.
- E. Apply appropriate access specifiers and encapsulation to prevent direct modification of sensitive financial data.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class DigitalWallet {
4     int walletID;
5     string name, mobile;
6     double balance;
7 public:
8     static int count;
9     void createWallet() {
10         cin >> walletID >> name >> mobile >> balance;
11         count++;
12     }
13     void deposit(double amount) {
14         balance += amount;
15     }
```

Input

```
2
101 Ravi 9876543210
2000 1000
102 Priya 9876501234
.....
```

Output

Q4 Smart Agriculture Crop Monitoring System

Q1

AI-Based Student Performance
Analytics System
10 marks

Q2

Digital Healthcare Patient
Monitoring System
10 marks

Q3

Digital Wallet and FinTech
Transaction System
10 marks

Q4

Smart Agriculture Crop
Monitoring System
10 marks

Q5

Smart City IoT Energy
Management System
10 marks

5/5 answered

Smart Agriculture Crop Monitoring System Scenario: IoT-Based Agriculture. An agricultural research organization is developing a Smart Agriculture System to monitor crop fields using sensor data and identify whether crops require irrigation. The system should be designed to handle the following requirements:

- A. Design a class CropField with attributes such as Field ID, Crop Name, Soil Moisture, Temperature, Humidity, and Irrigation Status.
- B. Implement member functions to input sensor readings, evaluate soil conditions, determine irrigation requirements, and update the irrigation status.
- C. Use an array of objects to maintain information about multiple crop fields.
- D. Include a static member to count the total number of monitored fields.
- E. Ensure proper encapsulation and modular object-oriented design suitable for an IoT-based agricultural system.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class CropField {
4     int fieldID;
5     string cropName;
6     float soilMoisture, temperature, humidity;
7     string irrigationStatus;
8 public:
9     static int count;
10    void input() {
11        cin >> fieldID >> cropName;
12        cin >> soilMoisture >> temperature >> humidity;
13        count++;
14    }
```

Input

```
2
101 Rice
102 Wheat
```

Output

Q5 Smart City IoT Energy Management System

Q1
AI-Based Student Performance
Analytics System
10 marks

Q2
Digital Healthcare Patient
Monitoring System
10 marks

Q3
Digital Wallet and FinTech
Transaction System
10 marks

Q4
Smart Agriculture Crop
Monitoring System
10 marks

Q5
Smart City IoT Energy
Management System
10 marks

5/5 answered

Smart City IoT Energy Management System Scenario: Sustainable Smart City. A smart-city research project aims to develop a Smart City IoT Energy Management System for monitoring electricity consumption in different buildings and identifying buildings with high energy consumption.

A. Create a class Building with attributes such as Building ID, Building Name, Building Type, Energy Consumption, and Energy Status.

B. Implement member functions to input building details, calculate energy status based on consumption, and display building information.

C. Use an array of objects to manage multiple buildings in the smart-city system.

D. Include a static member to maintain the total number of buildings monitored.

E. Apply encapsulation, data hiding, and structured C++ object-oriented programming principles in the implementation.

Your Code

```
1 #include <iostream>
2 using namespace std;
3 class Building {
4     int id;
5     string name, type, status;
6     float consumption;
7 public:
8     static int count;
9     void input() {
10         cin >> id >> name >> type >> consumption;
11         count++;
12     }
13     void checkStatus() {
14         if (consumption > 1000)
```

Input

2
101 Office Commercial
102 Home Residential

Output



Search

